Band-1 Hadamard Order 6676 — Miyamoto (Paley-I H_1668 + C_1670 over prime 1669); Miyamoto-only Bonus (NOT in SPK); Gap 2052 Still OPEN

H_1668 Paley-I PASS; C_1670 PASS; H_6676 Miyamoto PASS; Paley-I/II/Kronecker FAIL → Miyamoto-only; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-15

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_6628_Miyamoto`; `2026-09-15_Band1_Hadamard_6484_Miyamoto`; `2026-09-15_Band1_Hadamard_6436_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_6676_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 1668 and 6676 only** (verified max $|HH^\top - nI| = 0$), via Miyamoto with Paley-I seed H_{1668} = Paley-I$(q = 1667)$ and symmetric Paley conference C_{1670} over prime $q = 1669 \equiv 1 \pmod 4$. Classical SPK paths for 6676: Paley-I **FAIL**s ($q = 6675 = 3 \cdot 5^2 \cdot 89$ not prime); Paley-II **FAIL**s ($q = 3337 = 47 \cdot 71$ not prime); Kronecker **FAIL**s (4×1669). Therefore order **6676 is a Miyamoto-only bonus (NOT in SPK)**. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. `clay_closed = false`. Prior skip $q = 1637 \to n = 6548$ retained; no intervening conference-q skip between 1657 and 1669 ($1661 = 11 \cdot 151$, $1665 = 3^2 \cdot 5 \cdot 37$). Prior SPK seeds **6628** / **1656** / **1620** / **1608** / **6436** retained; Miyamoto-only bonuses **6484/6404/6388/6196/…** and **6676** excluded from ONBOX_SEEDS.

Abstract

Finite constructions of orders **1668** and **6676**.

- H_{1668} = Paley-I ($q = 1667$) — preferred Miyamoto seed
- C_{1670} = Paley conference over prime 1669
- H_{6676} = Miyamoto(H_{1668}, C_{1670})
- Paley-I / Paley-II for 6676: **SKIP**

SPK: 6676 Miyamoto-only bonus (NOT in SPK). Seed 1668 enters SPK via Paley-I. Prior SPK **6628** / **1656** / **1620** / **1608** / **6436** retained in ONBOX.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

```
python3 verify_hadamard_6676.py
```

Results. overall_pass=true; units 89/89; wall 242.9883s; all max|HH^T-nI|=0.

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Arithmetic

- Target $n = 6676 = 4 \cdot 1669$; 1669 prime $\equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{1668} and C_{1670} (Paley conference over prime 1669).
- Seed: Paley-I $q = 1667$ (prime $\equiv 3 \pmod 4$) → H_{1668}.
- SPK: Paley-I $q = 6675$ **FAIL**; Paley-II $q = 3337$ **FAIL**; Kronecker **FAIL** → **Miyamoto-only**.
- Prior skip: $q = 1637 \rightarrow n = 6548$ (no easy seed). No intervening skip 1657→1669.

Constructions (verified)

Object	Method	max|HH^T-nI|	Wall (s)
H_1668 Paley-I	Paley-I q=1667	0	3.3585
C_1670	Paley conf q=1669	0	3.2684
H_6676 Miyamoto	Miyamoto(H_1668 Paley-I, C_1670)	0	233.0674
H_6676 Paley-I	SKIP q=6675=3·5²·89 not prime	—	—
H_6676 Paley-II	SKIP q=3337=47·71 not prime	—	—

SPK closure note

Seeds include prior SPK ships through **6628** lineage (**6628** Paley-II; **1656** Double Paley-I; **1620** Paley-I/II; **1608** Paley-I; **6436** Paley-II; **NOT** 6484/6404/6388/6196/6676) + ver-

ified **1668** (Paley-I); **exclude** Miyamoto-only bonuses including **6676**/**6484**/**6404**/**6388**/**6196**. Known count 1392. Next constructive gap **2052** (OPEN).

Quantity	Value
6676 in SPK	**False** (Miyamoto-only bonus)
1668 in SPK	**True** (Paley-I q=1667)
6628 in SPK	**True** (prior Paley-II q=3313)
1656 in SPK	**True** (prior Double Paley-I)
1620 / 1608 / 6436 in SPK	**True** (prior)
Next gap	**2052** OPEN

Honesty

- `clay_closed = false`
- GATE-04 collection **LOCKED**
- Hadamard conjecture **OPEN**
- Gap **2052** still **OPEN** (method triage / non-finding ONLY)
- Claim level = finite constructions **1668+6676 ONLY**
- **6676 Miyamoto-only bonus (NOT in SPK)**

Next optional targets

- q=1681 → n=6724 need H_1680: Double(PaleyI half=840), Kronecker(4x420)
- q=1693 → n=6772 need H_1692: (no easy seed)
- q=1697 → n=6788 need H_1696: Kronecker(4x424)
- q=1709 → n=6836 need H_1708: PaleyII(q=853)
- q=1721 → n=6884 need H_1720: Double(PaleyI half=860)
- q=1733 → n=6932 need H_1732: (no easy seed)

Also: method-attack SPK gap **2052**; Track B privacy / Object 1 meter await; q=1613→n=6452 Paley-I q=6451 (possible SPK) if revisited.

Artifacts

- `verify_hadamard_6676.py`, `results.json`, `literature_ledger.json`
- `H1668_paleyI_pm1.npz`, `C1670_pm1.npz`, `H6676_miyamoto_pm1.npz`
- Shelf: `/workspace/DOWNLOAD_PDFs/`

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